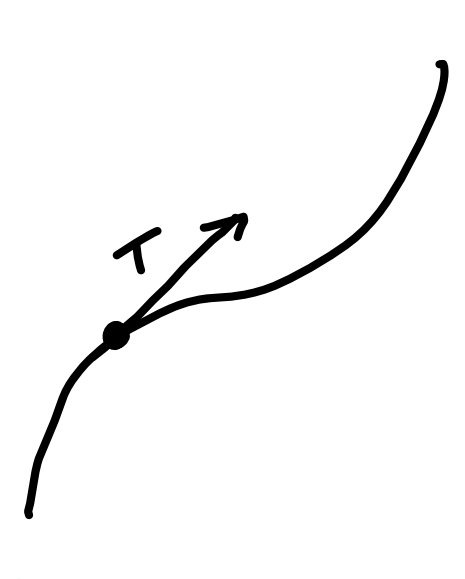
Tangent Vectors

Section 12.4

9-24-26

A) Unit Tangent Vector

$$T(t) = \frac{r'(t)}{\|r'(t)\|} = \frac{v(t)}{\|v(t)\|}$$



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ex) Find $T(t)$ and tangent line for $r(t) = \langle 2\cos t, \sin t, t \rangle$ at $(0, 1, \pi/2)$

$r'(t) = \langle -2\sin t, \cos t, 1 \rangle$

$\|r'(t)\| = \sqrt{4\sin^2 t + \cos^2 t + 1}$

$T(t) = \frac{1}{\sqrt{4\sin^2 t + \cos^2 t + 1}} \cdot r'(t)$

$r'(\frac{\pi}{2}) : \langle -2, 0, 1 \rangle$

$\|r'(\frac{\pi}{2})\| = \sqrt{5}$

$T(\frac{\pi}{2}) = \langle \frac{-2}{\sqrt{5}}, 0, \frac{1}{\sqrt{5}} \rangle$

B) Principal Unit Normal Vector

Normal vector ⊥ to T

There are a lot of vectors ⊥ to T

$\frac{d}{dt} T \cdot T = \frac{d}{dt} \#$

$T' \cdot T + T' \cdot T = 0$

$T' \cdot T = 0$

$$N(t) = \frac{T'(t)}{\|T'(t)\|}$$

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ex) $r(t) = \langle \sqrt{2}, e^t, -e^{-t} \rangle$ at $t = 0$

First Find $T(t)$

$r'(t) = \langle 0, e^t, e^{-t} \rangle$

$\|r'(t)\| = \sqrt{e^{2t} + \frac{1}{e^{2t}}} = \sqrt{\frac{e^{4t}+1}{e^{t}}}$

$T(t) = \langle 0, \frac{e^{2t}}{\sqrt{e^{4t}+1}}, \frac{1}{\sqrt{e^{4t}+1}} \rangle$

Then Find $T'(t)$

$$\left\langle 0, \frac{2e^{2t}\sqrt{e^4+1} - \frac{4e^{4t} \cdot e^{2t}}{2\sqrt{e^{4t}+1}}}{e^{4t}+1}, -\frac{14e^{4t}}{2(e^4+1)^{3/2}} \right\rangle$$

$$\left\langle 0, \frac{2e^{2t}}{(e^{4t}+1)^{3/2}}, \frac{-2e^{4t}}{(e^{4t}+1)^{3/2}} \right\rangle = \frac{2e^{2t}}{(e^{4t}+1)^{3/2}} \langle 0, 1, -e^{2t} \rangle$$

$T'(0) = \langle 0, \frac{2}{2^{3/2}}, -\frac{2}{2^{3/2}} \rangle$

$T'(0) = \langle 0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \rangle$

$\|T'(0)\| = 1$

$N(0) = \langle 0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \rangle$

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7)

$r'(t) = \langle 1, 2t, 1 \rangle$

$\|r'(t)\| = \sqrt{2+4t^2}$

$T(t) = \langle \frac{1}{\sqrt{2+4t^2}}, \frac{2t}{\sqrt{2+4t^2}}, \frac{1}{\sqrt{2+4t^2}} \rangle$

$r'(0) = \langle 1, 0, 1 \rangle$ $\|r'(0)\| = \sqrt{2}$

$T(0) = \langle \frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \rangle$

$x(t) = 0 + \frac{1}{\sqrt{2}}t$ $y(t) = 0$ $z(t) = 0 + \frac{1}{\sqrt{2}}t$

11)

$r(t) = \langle 2\cos t, 2\sin t, 4 \rangle$

$r'(t) = \langle -2\sin t, 2\cos t, 0 \rangle$

$\|r'(t)\| = \sqrt{4\sin^2 t + 4\cos t} = \sqrt{4} = 2$

$T(t) = \langle \frac{-2\sin t}{2}, \frac{2\cos t}{2}, 0 \rangle$

$r'(\frac{\pi}{4}) = \langle -\sqrt{2}, \sqrt{2}, 0 \rangle$

$T(\frac{\pi}{4}) = \langle -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0 \rangle$

$x(t) = \frac{\pi}{4} - \frac{\sqrt{2}}{2}t$ $y(t) = \frac{\pi}{4} + \frac{\sqrt{2}}{2}t$ $z(t) = \frac{\pi}{4}$

13)

$r(t) = t i + \frac{1}{2}t^2 j$ for $t = 2$

$r'(t) = \langle 1, t, 0 \rangle$

$\|r'(t)\| = \sqrt{t^2+1}$

$T(t) = \langle \frac{1}{\sqrt{t^2+1}}, \frac{t}{\sqrt{t^2+1}}, 0 \rangle$

$$T'(t) = \left\langle \frac{-2t}{2(t^2+1)^{3/2}}, \frac{1 \cdot \sqrt{t^2+1} - \frac{2t \cdot t}{2\sqrt{t^2+1}}}{(1+t^2)}, 0 \right\rangle$$

$T'(t) = \langle \frac{-t}{(1+t^2)^{3/2}}, \frac{1}{(1+t^2)^{3/2}}, 0 \rangle$

$T'(2) = \langle -\frac{1}{4}, \frac{1}{8}, 0 \rangle$

$\|T'(2)\| = \frac{1}{5}$

$N(2) = \langle -\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0 \rangle$

extra stuff:

$\|T'(t)\| = \frac{1}{(1+t^2)^{3/2}} \sqrt{1+t^2} = \frac{1}{1+t^2}$

$N(t) = \langle \frac{-t}{\sqrt{1+t^2}}, \frac{1}{\sqrt{1+t^2}}, 0 \rangle$

15)

$r(t) = \langle \ln t, t+1, 0 \rangle$

$r'(t) = \langle \frac{1}{t}, 1, 0 \rangle$

$\|r'(t)\| = \sqrt{1 + \frac{1}{t^2}} = \sqrt{\frac{t^2+1}{t^2}}$

$T(t) = \langle \frac{1}{\sqrt{t^2+1}}, \frac{t}{\sqrt{t^2+1}}, 0 \rangle$

$T'(t) = \langle -\frac{t}{(t^2+1)^{3/2}}, \frac{1}{(t^2+1)^{3/2}}, 0 \rangle$

$T'(2) = \langle -2, 1, 0 \rangle$

$\|T'(2)\| = \sqrt{5} \cdot 5^{-3/2} = \frac{1}{5}$

$N(2) = \langle -\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0 \rangle$

19)

$r(t) = \langle 6\cos t, 6\sin t, 1 \rangle$

$r'(t) = \langle -6\sin t, 6\cos t, 0 \rangle$

$\|r'(t)\| = \sqrt{36\sin^2 t + 36\cos^2 t} = \sqrt{36} = 6$

$T(t) = \langle -\sin t, \cos t, 0 \rangle$

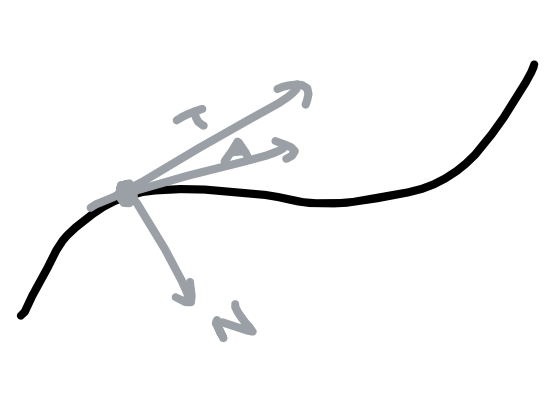
$T'(t) = \langle -\cos t, -\sin t, 0 \rangle$

$T'(\frac{3\pi}{4}) = \langle \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0 \rangle$

$\|T'(\frac{3\pi}{4})\| = \sqrt{\frac{4}{4}} = 1$

$N(\frac{3\pi}{4}) = \langle \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0 \rangle$

C) Tangental and Normal Components of Acceleration



$r(t) = \langle \quad \rangle$

$v(t) = r'(t) = \langle \quad \rangle$

$a(t) = r''(t) = \langle \quad \rangle$

$a = a_T T + a_N N$ $T(t) = \frac{v(t)}{\|v(t)\|} = \frac{r'(t)}{\|r'(t)\|}$

Centripetal Component of Acceleration

$$\frac{\|a \times N\|}{\|v\|} \cdot \vec{N}$$

$a_T = \frac{a \cdot v}{\|v\|}$ $a_N = \frac{\|a \times v\|}{\|v\|}$

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ex) $r(t) = \langle t^2, t^2, t^3 \rangle$

find a_N and a_T

$a_T = \frac{a \cdot v}{\|v\|}$ $a_N = \frac{\|a \times v\|}{\|v\|}$

$r(t) = \langle t^2, t^2, t^3 \rangle$

$r'(t) = \langle 2t, 2t, 3t^2 \rangle$

$r''(t) = \langle 2, 2, 6t \rangle$

$a_T = \frac{8t + 18t^3}{\sqrt{8t^2+9t^4}} = \frac{8+18t^2}{\sqrt{8+9t^2}}$

$a_N = \frac{6t^2\sqrt{2}}{\sqrt{8+9t^2}}$

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23)

$T(t) = \frac{v(t)}{\|v(t)\|} = \frac{\langle 1-3t^2, 4t, 0 \rangle}{\sqrt{9t^4+10t^2+1}}$

$a_T = \frac{14}{\sqrt{5}}$

$a_N = \frac{8}{\sqrt{5}}$

$N(t)$

$r(t) = \langle t - t^3, 2t^2, 0 \rangle$

$r'(t) = \langle 1-3t^2, 4t, 0 \rangle$

$r''(t) = \langle -6t, 4, 0 \rangle$

$$\overset{\vec{a}}{\langle -6, 4, 0 \rangle} = \overset{a_T \cdot \vec{T}}{\frac{14}{\sqrt{5}} \langle -\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0 \rangle} + \overset{a_N \cdot \vec{N}}{\frac{8}{\sqrt{5}} \langle a, b, c \rangle}$$

24)

$v(t) = \langle$

27)

$v(t) = e^t(\cos t - \sin t)i + e^t(\cos t + \sin t)j$

$a(t) = e^t(-2\sin t)i + e^t(2\cos t)j$

$a_T = \frac{a \cdot v}{\|v\|} = \sqrt{2e^{\pi}}$

$a_N = \frac{\|a \times v\|}{\|v\|} = \sqrt{2e^{\pi}}$

$v(\frac{\pi}{2}) = \langle -e^{\pi/2}, e^{\pi/2}, 0 \rangle$

$\|v(\pi/2)\| = \sqrt{2e^{\pi}}$

$T(\frac{\pi}{2}) = \langle -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \rangle$

42)

$r(t) = e^t i + 2t j + e^{-t} k$ $t = 0$

$r(t) = \langle e^t, 2t, e^{-t} \rangle$

$v(t) = r'(t) = \langle e^t, 2, -e^{-t} \rangle$

$a(t) = r''(t) = \langle e^t, 0, e^{-t} \rangle$

$$a_T = \frac{e^{2t} - e^{-2t}}{\sqrt{4 + e^{2t} + e^{-2t}}} = \frac{e^{3t} - \frac{1}{e^t}}{\sqrt{e^{4t} + 4e^{2t} + 1}}$$

$a \times v = \langle 2e^{-t}, -2, -2e^t \rangle$

$\|a \times v\| = \sqrt{4e^{-2t} + 4 + 4e^{2t}}$

$$a_N = \frac{\sqrt{4e^{-2t} + 4 + 4e^{2t}}}{\sqrt{4 + e^{2t} + e^{-2t}}}$$

$T = \frac{1}{\sqrt{4 + e^{2t} + e^{-2t}}} \langle e^t, 2, -e^{-t} \rangle$

$$\langle e^t, 0, e^{-t} \rangle = \frac{e^{2t} - e^{-2t}}{4 + e^{2t} + e^{-2t}} \langle e^t, 2, -e^{-t} \rangle + \frac{\sqrt{4e^{-2t} + 4 + 4e^{2t}}}{\sqrt{4 + e^{2t} + e^{-2t}}} (N)$$

@ $t = 0$

$\langle 1, 0, 1 \rangle = 0 + \sqrt{2}(N)$

$N = \langle \frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \rangle$

52) Find the Binormal vector

$r(t) = \langle t, t^2, \frac{t^3}{3} \rangle$

$v(t) = r'(t) = \langle 1, 2t, t^2 \rangle$

$a(t) = r''(t) = \langle 0, 2, 2t \rangle$

$a_T = \frac{4t + 2t^3}{\sqrt{t^4 + 4t^2 + 1}}$ $a \times v = \langle 2t^2, -2t, 2 \rangle$

$a_N = \frac{\sqrt{4t^4 + 4t^2 + 4}}{\sqrt{t^4 + 4t^2 + 1}}$

$$\langle 0, 2, 2t \rangle = \frac{4t + 2t^3}{t^4 + 4t^2 + 1} \langle 1, 2t, t^2 \rangle + \frac{\sqrt{4t^4 + 4t^2 + 4}}{\sqrt{t^4 + 4t^2 + 1}} (N)$$

$\langle 0, 2, 2 \rangle = \langle 1, 2, 1 \rangle + \sqrt{2}(N)$

$\langle -1, 0, 1 \rangle = \sqrt{2}(N)$

$T = \langle \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \rangle$

$N = \langle -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \rangle$

$T \times N = \langle \frac{2}{\sqrt{12}}, -\frac{2}{\sqrt{12}}, \frac{2}{\sqrt{12}} \rangle = \langle \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \rangle$

$B(t) = \langle \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

2243) Find $B(t)$ if $r(t) = \langle 2t, 2\sin t, 2\cos t \rangle$

$v(t) = r'(t) = \langle 2, 2\cos t, -2\sin t \rangle$

$a(t) = r''(t) = \langle 0, -2\sin t, -2\cos t \rangle$

$a_T = -4\sin t\cos t$